Daily Geometry Drill #5 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

Section A Rapid Recognition

1. **Answer:** 4

Method: Tangent length $= \sqrt{OP^2 - r^2} = \sqrt{25-9} = \sqrt{16} = 4$.

> **Investigate further:** Using $\sqrt{r^2 - OP^2}$ (the inside/outside sign flipped).

2. **Answer:** $\dfrac{3}{4}$

Method: $\tan\frac{\theta}{2} = \frac{r}{\sqrt{OP^2 - r^2}} = \frac{3}{4}$. The right triangle is exactly the tangent length geometry of A1.

> **Investigate further:** Writing $\sin\frac{\theta}{2}$'s $\frac{r}{OP} = \frac{3}{5}$ instead of the tangent ratio.

3. **Answer:** "$5x + 3y = 25$"

Method: The polar of (x_1, y_1) is $xx_1 + yy_1 = r^2$: $5x + 3y = 25$.

> **Investigate further:** Putting P's coordinates in as a point on the circle rather than coefficients of the line.

4. **Answer:** $5\sqrt{2}$

Method: $r^2 = 25$, director circle $x^2 + y^2 = 2r^2 = 50$, radius $\sqrt{50} = 5\sqrt{2}$.

> **Investigate further:** Doubling the radius r instead of the squared radius r^2.

5. **Answer:** $\sqrt{13}$

Method: $\sqrt{7^2 - 6^2} = \sqrt{49 - 36} = \sqrt{13}$.

> **Investigate further:** Subtracting r from OP (linear) instead of r^2 from OP^2.

6. **Answer:** $3\sqrt{3}$

Method: $\sin\frac{\theta}{2} = \frac{r}{OP}$ gives $\sin 30^\circ = \frac{r}{6}$, so $r = 3$, tangent $= \sqrt{36 - 9} = 3\sqrt{3}$.

> **Investigate further:** Using $\frac{\theta}{2} = 60^\circ$; the tangents make the full angle, halve it before the sine.

7. **Answer:** "$5x = 4$"

Method: Polar: $xx_1 + yy_1 = r^2$ with $(x_1, y_1) = (5, 0)$ gives $5x = 4$, the vertical chord halfway between the two symmetrical tangency points.

> **Investigate further:** Forgetting to include the yy_1 term that vanishes with $y_1 = 0$, or solving to $x = \frac{4}{5}$ when the polar itself is the answer.

8. **Answer:** 13

Method: $OP = \sqrt{12^2 + 5^2} = 13$.

> **Investigate further:** Adding $12 + 5 = 17$ or subtracting to 7.

9. **Answer:** $\dfrac{10\sqrt{3}}{3}$

Method: Half the angle is $60°$; $\sin 60° = \frac{5}{OP}$ so $OP = \frac{5}{\sqrt{3}/2} = \frac{10\sqrt{3}}{3}$.

> **Investigate further:** Taking $\frac{\theta}{2} = 120°$, or inverting the sine ratio.

10. **Answer:** ”$(4, 4)$”

Method: Compare $xx_1 + yy_1 = 16$ with $x + y = 4$: multiply the given line by 4 to get $4x + 4y = 16$, so $(x_1, y_1) = (4, 4)$.

> **Investigate further:** Reading the coefficients directly gives $(1, 1)$; the constant $r^2 = 16$ forces the scale 4.

Section B Manipulation Drills

1. **Answer:** B) 4

Method: $OP = \sqrt{9 + 16} = 5$, $r = 3$, tangent $= \sqrt{25 - 9} = 4$.

> **Investigate further:** Quoting $OP = 5$ (option C); the question asks for the tangent itself.

2. **Answer:** C) $60°$

Method: $\sin \frac{\theta}{2} = \frac{r}{OP} = \frac{3}{6} = \frac{1}{2}$, so $\frac{\theta}{2} = 30°$ and $\theta = 60°$.

> **Investigate further:** Answering $30°$, the half-angle, without doubling.

3. **Answer:** ”$3x + 5y = 25$”

Method: Polar: $xx_1 + yy_1 = 25$ gives $3x + 5y = 25$.

> **Investigate further:** Swapping x_1, y_1 with r^2 and writing $3x + 5y = r^2$ with the wrong constant.

4. **Answer:** B) $2\sqrt{2}$

Method: Directrix circle $x^2 + y^2 = 2r^2 = 8$, radius $\sqrt{8} = 2\sqrt{2}$.

> **Investigate further:** Answering the circle radius 2 (option A) instead of the director radius.

5. **Answer:** C) 2

Method: $\sqrt{1^2 + 2^2 - 1^2} = \sqrt{4} = 2$.

> **Investigate further:** Quoting $\sqrt{OP^2} = \sqrt{5}$ and ignoring the $r^2 = 1$ subtraction.

6. **Answer:** C) 13

Method: $OP = \sqrt{12^2 + 5^2} = 13$.

> **Investigate further:** Adding the segments (17) rather than Pythagoras.

7. **Answer:** B) 4

Method: Polar of $(a, 0)$ is $ax = 16$, i.e. $x = \frac{16}{a}$; setting this equal to 4 gives $a = 4$.

> **Investigate further:** Quoting $\frac{16}{4} = 4$ the wrong way round, or answering $a = 16$ as if no division happened.

8. **Answer:** 5

Method: Perpendicular tangents put P on the director circle: $OP^2 = 2r^2 = 50$; tangent $= \sqrt{50 - 25} = 5$.

> **Investigate further:** Answering the director circle's radius $5\sqrt{2}$.

9. **Answer:** A) 4

Method: Tangents of slope m to $x^2 + y^2 = r^2$ are $y = mx \pm r\sqrt{1 + m^2}$; here $r = \sqrt{8}$ so $c = \sqrt{8}\sqrt{2} = 4$.

> **Investigate further:** Quoting $r = \sqrt{8}$ itself as the intercept.

10. **Answer:** 8

Method: $r^2 = OP^2 - (\text{tangent})^2 = 100 - 36 = 64$, $r = 8$.

> **Investigate further:** Subtracting linearly $(10 - 6 = 4)$.

Section C Substitution & Structure

1. **Answer:** A) 4

Method: $OP = \sqrt{7 + 9} = 4$.

> **Investigate further:** Taking OP as the sum of the two squared legs, $\sqrt{16} = 4$-style thinking that skips the square root of the sum.

2. **Answer:** A) $x = 2$

Method: Equal powers: $x^2 + y^2 - 4 = (x - 6)^2 + y^2 - 16$, which cancels to $-4 = -12x + 20$, i.e. $x = 2$. The circles are externally tangent at $(2, 0)$.

> **Investigate further:** Subtracting the radii or reading the gap 6 directly; the radical axis sits at the tangent point $x = 2$.

3. **Answer:** B) $\dfrac{4}{5}$

Method: Distance $= \frac{|4\cdot 0 + 3\cdot 0 - 4|}{\sqrt{4^2 + 3^2}} = \frac{4}{5}$.

> **Investigate further:** Quoting $\frac{25}{5} = 5$ (pre-reduction) or flipping the fraction.

4. **Answer:** A) $2\sqrt{6}$

Method: Centre $C = (3, 4)$, $PC = \sqrt{3^2 + 4^2} = 5$, $r = 1$; tangent $= \sqrt{25 - 1} = \sqrt{24} = 2\sqrt{6}$.

> **Investigate further:** Reading the centre from the raw coefficients instead of halving them.

5. **Answer:** A) $3\sqrt{2}$

Method: Director circle of radius-3 circle has radius $3\sqrt{2}$.

> **Investigate further:** Doubling the radius to 6 (option B) or squaring it to 18 (option C).

6. **Answer:** A) $3\sqrt{11}$

Method: $OP = \sqrt{8^2 + 6^2} = 10$; tangent $= \sqrt{100 - 1} = \sqrt{99} = 3\sqrt{11}$.

> **Investigate further:** Answering $\sqrt{99}$ unsimplified (option B) or forgetting $r^2 = 1$.

7. **Answer:** B) 8

Method: The tangent segments from a vertex are $s - a$. Here $s = 21$ and the opposite side is 13, so the segments are $21 - 13 = 8$.

> **Investigate further:** Taking s itself or the side 13.

8. **Answer:** A) $\dfrac{\sqrt{5}}{5}$

Method: Distance $= \frac{1}{\sqrt{2^2+1^2}} = \frac{1}{\sqrt{5}} = \frac{\sqrt{5}}{5}$.

> **Investigate further:** Answering the coefficient ratio $\frac{1}{2}$ or skipping the normalization.

Section D Challenge Ramp

1. **Answer:** C) $60°$

Method: $\sin\frac{\theta}{2} = \frac{3}{6} = \frac{1}{2}$ gives $\frac{\theta}{2} = 30°$, so $\theta = 60°$.

> **Investigate further:** Leaving the half-angle $30°$ undoubled.

2. **Answer:** A) 10

Method: $OP = \sqrt{(5\sqrt{3})^2 + 5^2} = \sqrt{75 + 25} = \sqrt{100} = 10$.

> **Investigate further:** Squaring the tangent length incorrectly or adding $5 + 5\sqrt{3}$.

3. **Answer:** A) $3\sqrt{2}$

Method: Distance of $x + y = 3$ from the origin is $\frac{3}{\sqrt{2}}$; half-chord $= \sqrt{9 - \frac{9}{2}} = \frac{3}{\sqrt{2}}$, so the chord is $\frac{6}{\sqrt{2}} = 3\sqrt{2}$.

> **Investigate further:** Quoting the distance $\frac{3}{\sqrt{2}}$ (option C) as the chord length.

4. **Answer:** B) $5\sqrt{3}$

Method: $\sin 30° = \frac{r}{10} \Rightarrow r = 5$; tangent $= \sqrt{100-25} = 5\sqrt{3}$.

> **Investigate further:** Quoting $r = 5$ (option C) or $OP = 10$ (option A).

5. **Answer:** A) $\dfrac{27}{5}$

Method: Write $9x + 12y = 81$ as $3x + 4y = 27$; distance $= \frac{27}{\sqrt{9+16}} = \frac{27}{5}$.

> **Investigate further:** Using 81 unreduced (distance 81/15), or dividing the wrong way.

Top 5 Patterns

- **Tangent length is one Pythagorean step:** $\sqrt{OP^2 - r^2}$**.** Locate OP and r, square, subtract. *(see A1, A5, A8, B1, B5, B6, B10, C4, C6, D2)*
- **The polar of** (x_1, y_1) **is** $xx_1 + yy_1 = r^2$**.** The same r^2 on the right keeps the chord of contact honest. *(see A3, A7, A10, B3, B7, C3, C8, D3, D5)*
- $\sin\frac{\theta}{2} = \frac{r}{OP}$ **sizes the double-tangent wedge.** Halve the angle before you take the sine. *(see A2, A6, A9, B2, D1, D4)*
- **Perpendicular tangents define the director circle** $x^2 + y^2 = 2r^2$**.** The doubling happens on r^2, so the radius multiplies by $\sqrt{2}$. *(see A4, B4, B8, C5)*
- **Polar, half-chord and distance form one triangle.** The chord of contact is bisected by the perpendicular from the centre. *(see C3, C8, D3, D5, A3)*

Common Traps

- **Subtracting radii instead of squared radii.** $OP^2 = r^2 + \ell^2$; the null case is 7^2, not 7, minus 6^2. *(see A5, B10, D2)*
- **Answering the half-angle.** The tangents make θ; $\sin\frac{\theta}{2}$ is half of it. *(see B2, D1, D4)*
- **Reading polar coefficients as a point.** $5x + 3y = 25$ is a line; only its coefficients come from P. *(see A3, B3, A10)*
- **Bumping the director circle radius by one** $\sqrt{2}$ **too many or too few.** Doubling r^2 changes the radius by $\sqrt{2}$, not by 2. *(see A4, B4, C5)*
- **Leaving the polar unreduced.** The chord $9x + 12y = 81$ asks for the distance to the simplified $3x + 4y = 27$. *(see D5, C8, C3)*

"Every view of a tangent is the same right triangle: OP hypoteuse, r one leg, the tangent the other. The polar just points it."

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths